Pierre Houédry

4 cours d'Helsinki
35200 Rennes
France
+33687777285
p.houedry@gmail.com
pierre-houedry

Work Experience

2023-2024 **Research Engineer**, *University of Rennes*

Under the supervision of Valérie Monbet. Working on different Machine Learning projects:

1. Identification of interactions between bacterias of the gut microbiota playing a role in the irritable bowel syndrome (using data science and machine learning).
2. Prediction of rate of pollens in the air from photoelectric signals in collaboration with the startup LifyAir (using deep learning).

Nov. 2023 **Industrial Problem Solving Workshop**, *Marseille*

Organized by the french agency for Mathematics in interaction with industries and society (AMIES).

Education

2020-2023 **PhD in Mathematics**, *University of Caen Normandie & University of Rennes*

Twisted differential operators in several variables.
Under the supervision of Bernard Le Stum and Jérôme Poineau.

2019-2020 **MSc in Cryptography**, *University of Rennes*, with highest honour

Master thesis: *Introduction to p-adic q-differences equations* under the supervision of Bernard Le Stum.

2016-2019 **MSc in Mathematics**, *École Normale Supérieure of Rennes*, with high honours

Master thesis: *Pairing of elliptic curves for cryptography* under the supervision of Christophe Ritzenthaler.

Programming

Level	Skill	Years	Comment
■■■■□	Python	4	*Machine Learning projects using pandas, scikit-learn and tensorflow libraries: Customly regularized LSTM Deep AutoEncoders for anomaly detection in multivariate time series. Functional Data Analysis for nuclear experiments. Statistical analysis for biological high dimensional and low sample size data. Currently learning PyTorch. Mathematical numerical tools.*
■■■□□	Sage	3	*Implementation of cryptographic algorithms.*

Academic Experience

2023 **Master thesis supervisor**, *École Normale Supérieure of Rennes*
Supervision of the 1st year master's thesis of two students.

2020-2023 **Seminar supervisor**, *University of Rennes*
Arithmetic, Analysis and Probabilities for 1st year students. Holomorphic functions for 3rd year students. Formal methods for 2nd year students.

July-Aug. 2022 **Lecturer**, *The University of Tokyo*
By invitation of Atsushi Shiho and funding of the Japanese Society for Promotion of Sciences. Lecturer in a research group.

Sept.-Oct. 2021 **Research stay**, *Universidad Autónoma de Madrid*
By invitation of Adolfo Quirós.

May-July 2020 **Research Internship**, *IRMAR in Rennes*
Under the supervision of Bernard Le Stum.

Apr.-Aug. 2019 **Research internship**, *INRIA in Rennes*
On floating-point arithmetic for sampling discrete gaussian distributions.
Under the supervision of Adeline Roux-Langois and Mohamed Sabt.

July-Aug. 2018 **Research Internship**, *The University of Tokyo*
Local class field theory and absolute Galois groups. Under the supervision of Atsushi Shiho.

June-July 2016 **LMNO in Caen**
Differential Galois theory and Schanuel's conjecture. Under the supervision of Jérôme Poineau.

June-July 2014 **Research Internship**, *IRMAR in Rennes*
Special functions and Riemann hypothesis. Under the supervision of Vincent Mineo-Kleiner.

Publications and Talks

2023 **Twisted differential operators in several variables**
arXiv:2303.07756

2023 **Twisted calculus in several variables**
arXiv:2309.13277

Nov. 2022 **Institut Fourier in Grenoble**, *Number theory seminar*

July 2022 **The University of Tokyo**, *Arithmetic Geometry seminar*

June 2022 **IRMAR in Rennes**, *Arithmetic Geometry seminar*

Sept. 2021 **Universidad Autónoma de Madrid**, *Arithmetic Geometry seminar*

Oct. 2020 **LMNO in Caen**, *Number theory seminar*

Languages

Français	Mother tongue
Polonais	Fluent
Anglais	Fluent
Espagnol	Notions

Hobbies

Thai boxing (since 6 years) and art (drawing, painting, video editing and photography).